

Configure ASA Basic Settings and Firewall Using the CLI

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INTRODUCTION

This was an assignment on configuring basic settings and a firewall on a Cisco Adaptive Security Appliance (ASA) using the Command Line Interface (CLI) by using a set of commands that set up a devices essential and network security functions.

Methodology

Part 1: Verify Connectivity and Explore the ASA

Step 1: Verify connectivity.

The ASA is not currently configured. However, all routers, PCs, and the DMZ server are configured. Verify that PC-C can ping any router interface. PC-C is unable to ping the ASA, PC-B, or the DMZ server.

The screenshot displays the Cisco Packet Tracer interface. On the left, the 'Logical' tab is active, showing a network diagram with devices R1, R2, R3, ASA, and DMZ Server. Below the diagram is an 'Addressing Table'.

Device	Interface	IP Address
R1	G0/0	209.165.200.225
	S0/0/0 (DCE)	10.1.1.1
R2	S0/0/0	10.1.1.2
	S0/0/1 (DCE)	10.2.2.2
R3	G0/1	172.16.3.1
	S0/0/1	10.2.2.1
ASA	G1/1	209.165.200.226
	G1/2	192.168.1.1
	G1/3	192.168.2.1
DMZ Server	NIC	192.168.2.3

On the right, a 'PC-C' window is open, showing a Command Prompt. The prompt displays the results of ping tests from PC-C to various destinations:

```
Reply from 172.16.3.1: bytes=32 time=50ms TTL=255
Reply from 172.16.3.1: bytes=32 time<1ms TTL=255
Reply from 172.16.3.1: bytes=32 time<1ms TTL=255
Reply from 172.16.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.2.2.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 50ms, Average = 14ms

C:\>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:

Reply from 10.1.1.1: bytes=32 time=47ms TTL=253
Reply from 10.1.1.1: bytes=32 time=3ms TTL=253
Reply from 10.1.1.1: bytes=32 time=62ms TTL=253
Reply from 10.1.1.1: bytes=32 time=2ms TTL=253

Ping statistics for 10.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 52ms, Average = 26ms

C:\>ping 10.2.2.2

Pinging 10.2.2.2 with 32 bytes of data:

Reply from 10.2.2.2: bytes=32 time=24ms TTL=254
Reply from 10.2.2.2: bytes=32 time=17ms TTL=254
Reply from 10.2.2.2: bytes=32 time=17ms TTL=254
Reply from 10.2.2.2: bytes=32 time=19ms TTL=254

Ping statistics for 10.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 24ms, Average = 19ms

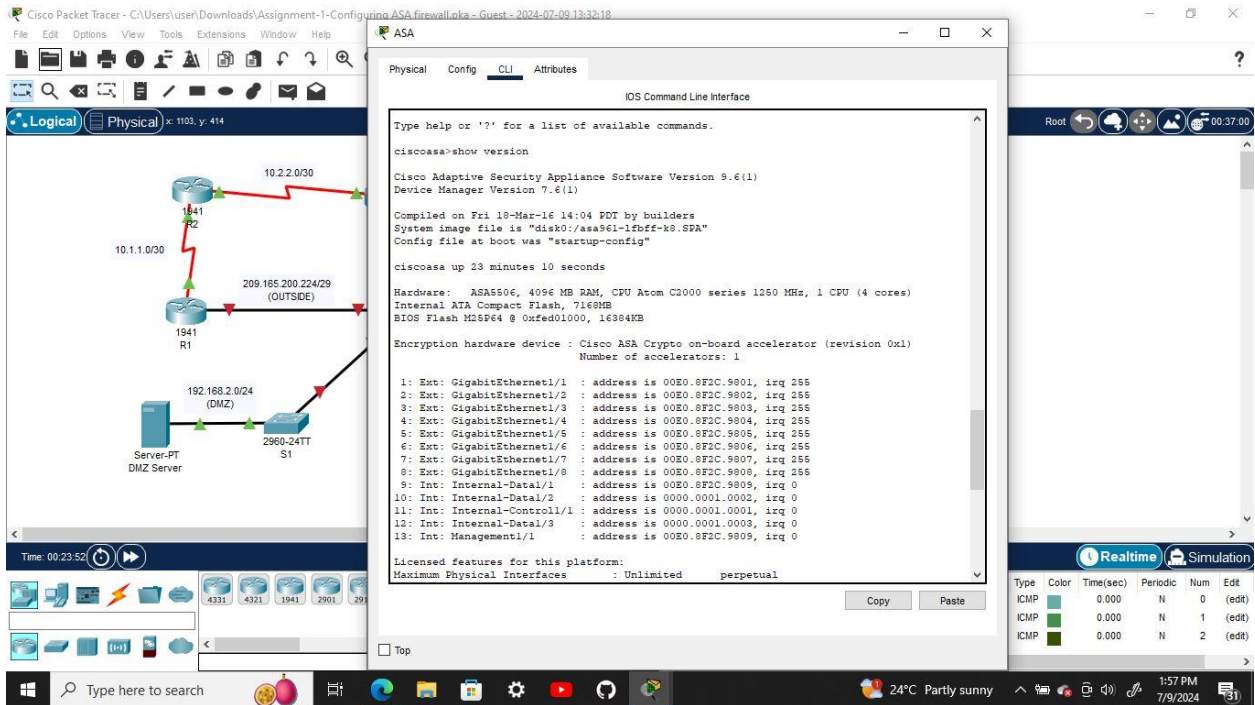
C:\>
```

At the bottom right, a 'Realtime' tab shows a table of network events:

Type	Color	Time(sec)	Periodic	Num	Edit
ICMP	Green	0.000	N	0	(edit)
ICMP	Green	0.000	N	1	(edit)
ICMP	Green	0.000	N	2	(edit)

Step 2: Determine the ASA version, interfaces, and license.

Use the `show version` command to determine various aspects of this ASA device.



Step 3: Determine the file system and contents of flash memory.

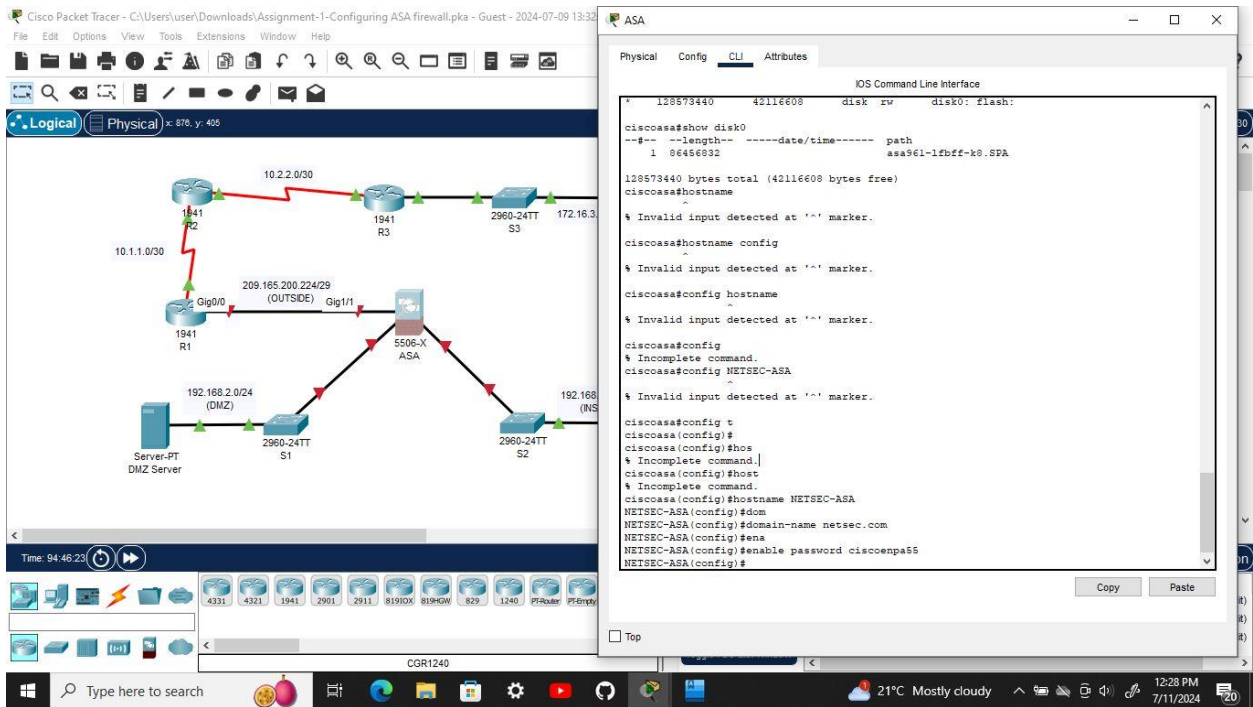
- Enter privileged EXEC mode. A password has not been set. Press **Enter** when prompted for a password.
- Use the **show file system** command to display the ASA file system and determine which prefixes are supported.
- Use the **show flash:** or **show disk0:** command to display the contents of flash memory.

Part 2: Configure ASA Settings and Interface Security Using the CLI

Tip: Many ASA CLI commands are similar to, if not the same, as those used with the Cisco IOS CLI. In addition, the process of moving between configuration modes and submodes is essentially the same.

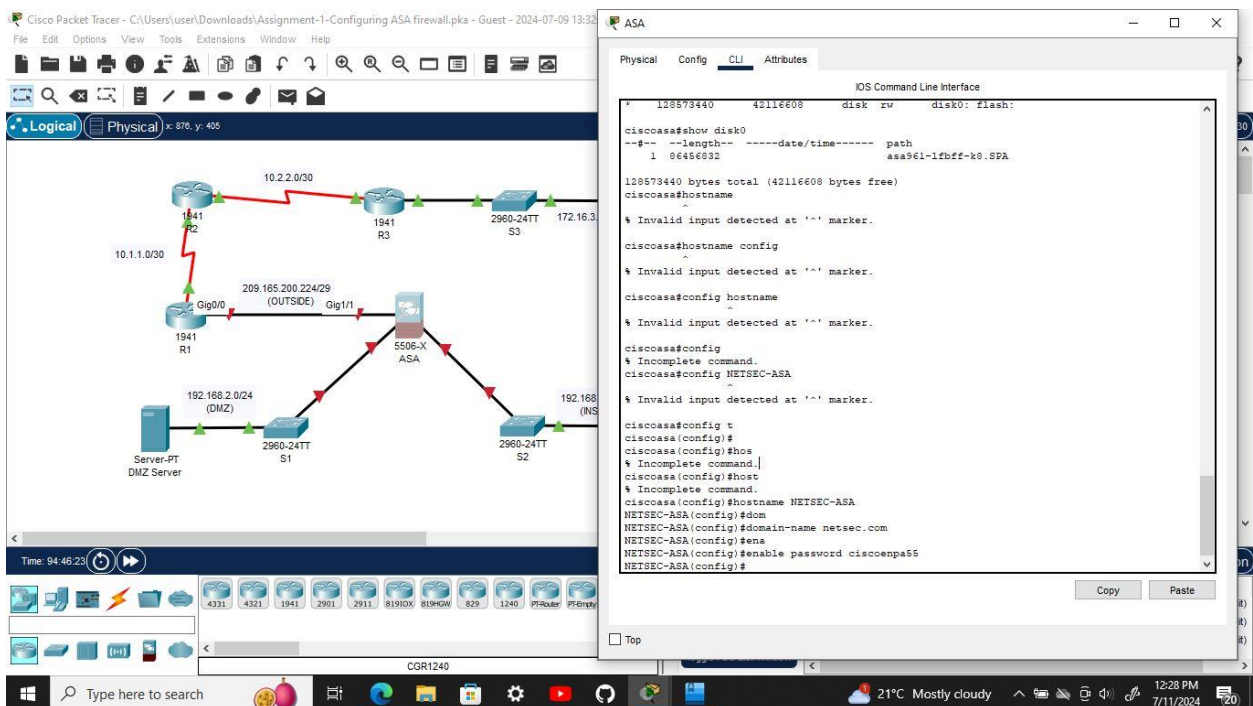
Step 1: Configure the hostname and domain name.

- Configure the ASA hostname as **NETSEC-ASA**.
- Configure the domain name as **netsec.com**.



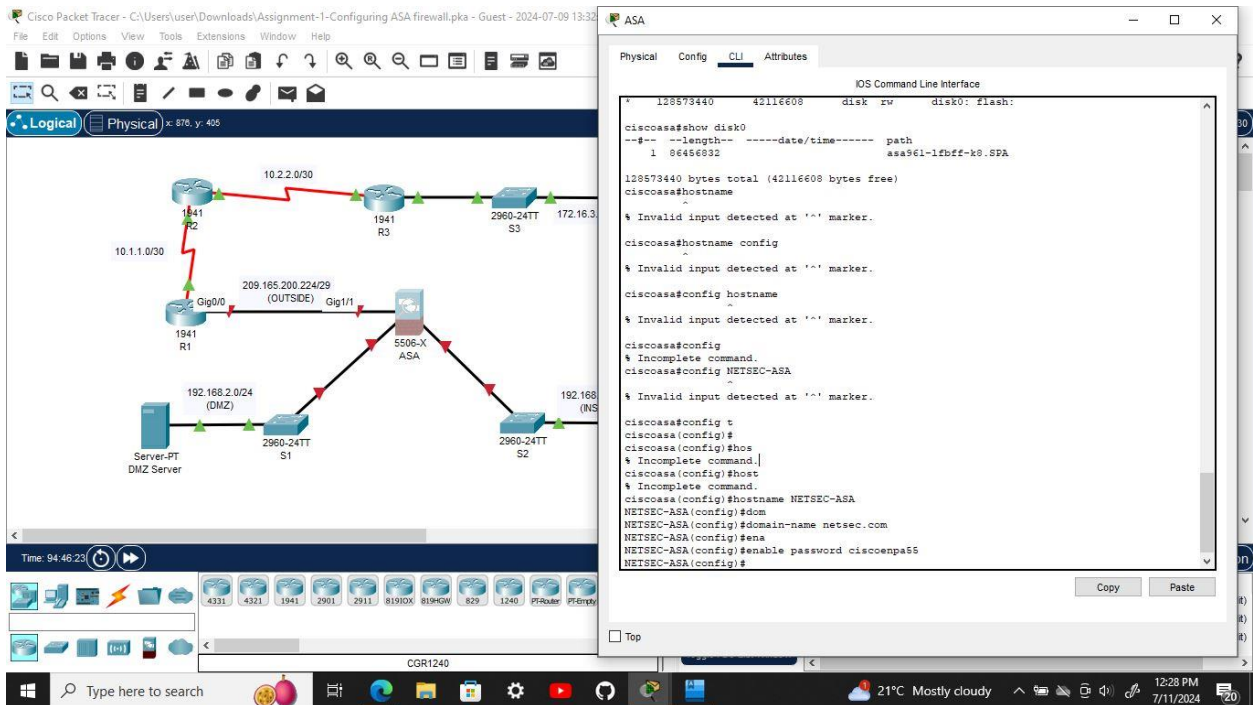
Step 2: Configure the enable mode password.

Use the **enable password** command to change the privileged EXEC mode password to **ciscoenpa55**.



Step 3: Set the date and time.

Use the **clock set** command to manually set the date and time (this step is not scored).



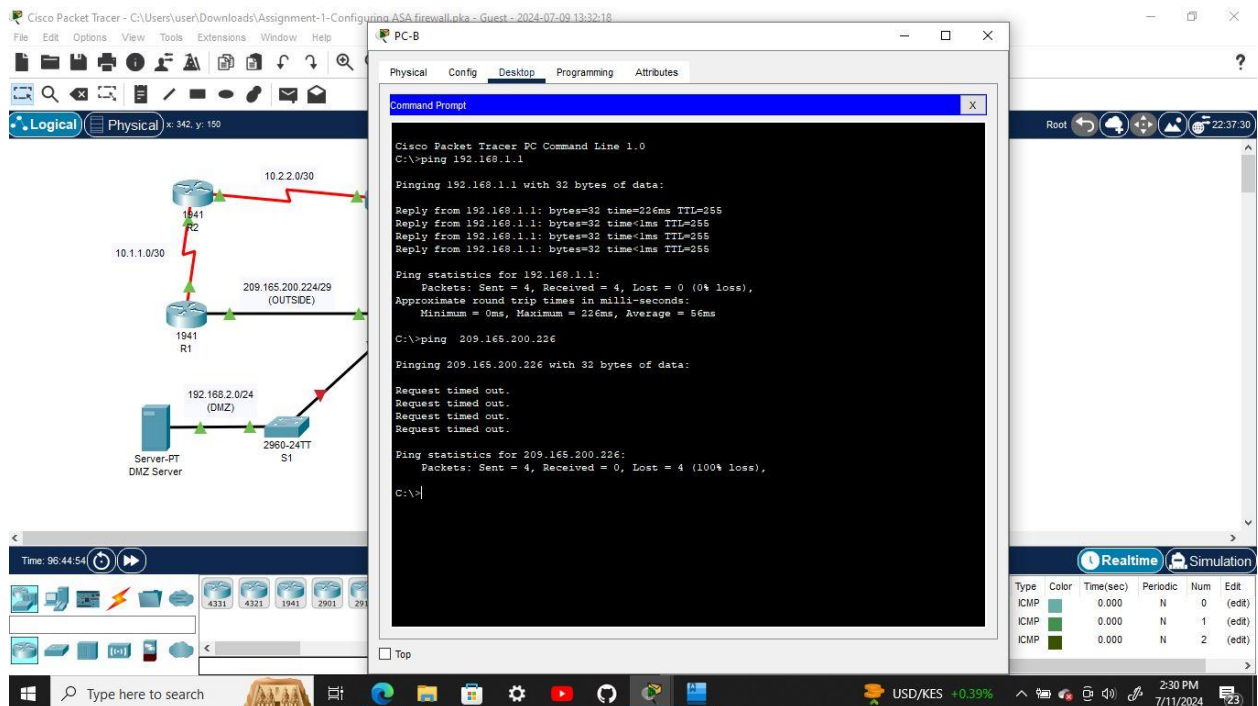
Step 4: Configure the INSIDE and OUTSIDE interfaces.

You will only configure the G1/1 (OUTSIDE) and G1/2 (INSIDE) interfaces at this time. The G1/3 (DMZ) interface will be configured in Part 5 of the activity.

- Create the G1/1 interface for the outside network (209.165.200.224/29), set the security level to the lowest setting of 0, and enable the interface.
- Configure the G1/2 interface for the inside network (192.168.1.0/24) and set the security level to the highest setting of 100 and enable the interface.

Step 5: Test connectivity to the ASA.

- You should be able to ping from PC-B to the ASA inside interface address (192.168.1.1). If the pings fail, troubleshoot the configuration as necessary.
- From PC-B, ping the G1/1 (OUTSIDE) interface at IP address 209.165.200.226. You should not be able to ping this address.

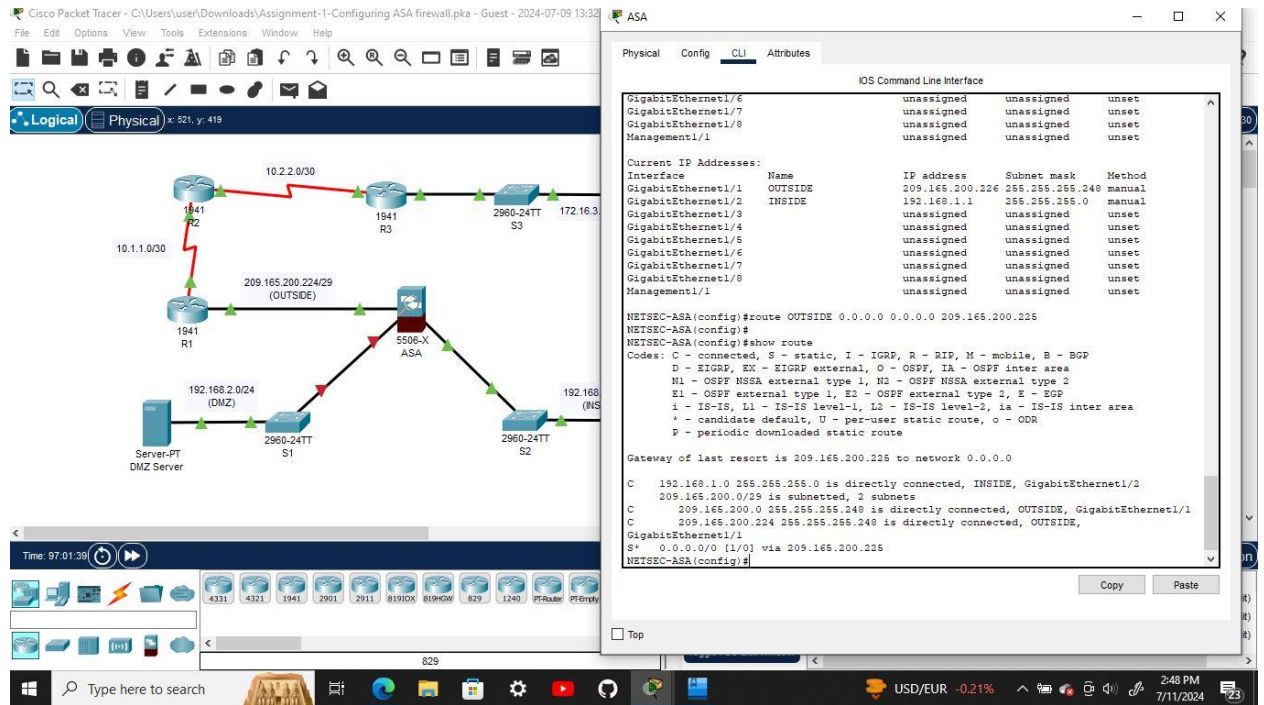


Part 3: Configure Routing, Address Translation, and Inspection Policy Using the CLI

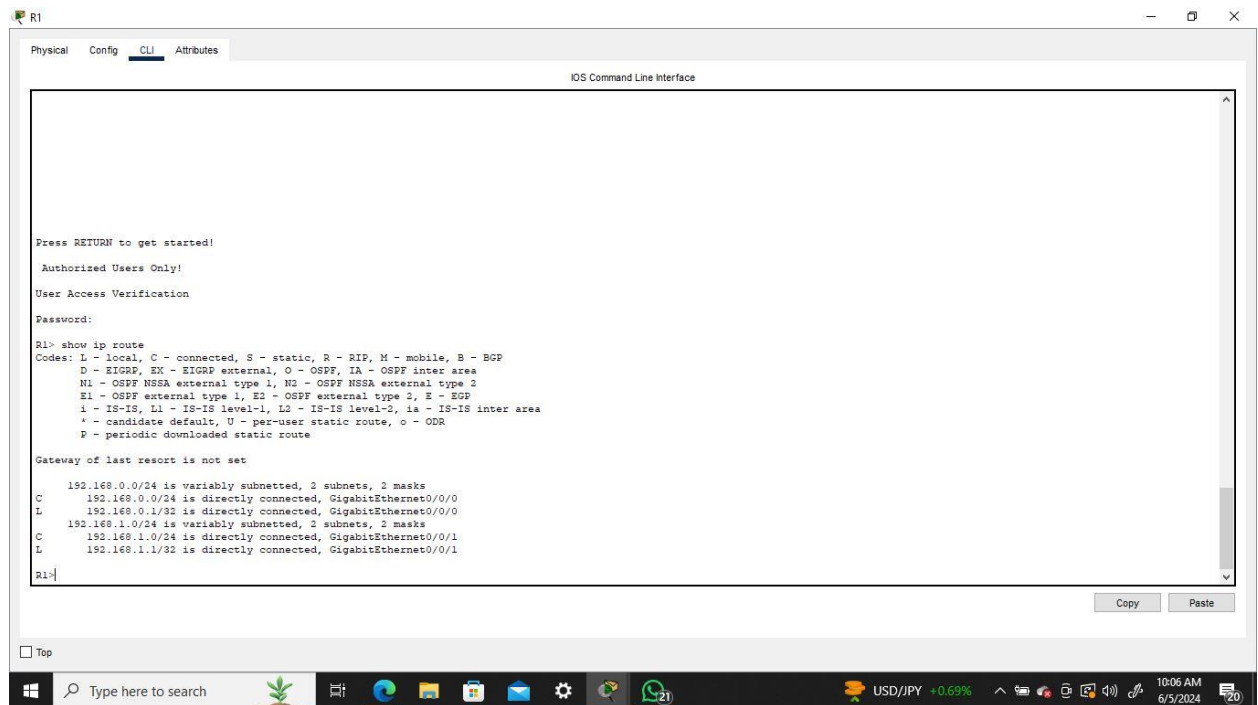
Step 1: Configure a static default route for the ASA.

Configure a default static route on the ASA OUTSIDE interface to enable the ASA to reach external networks.

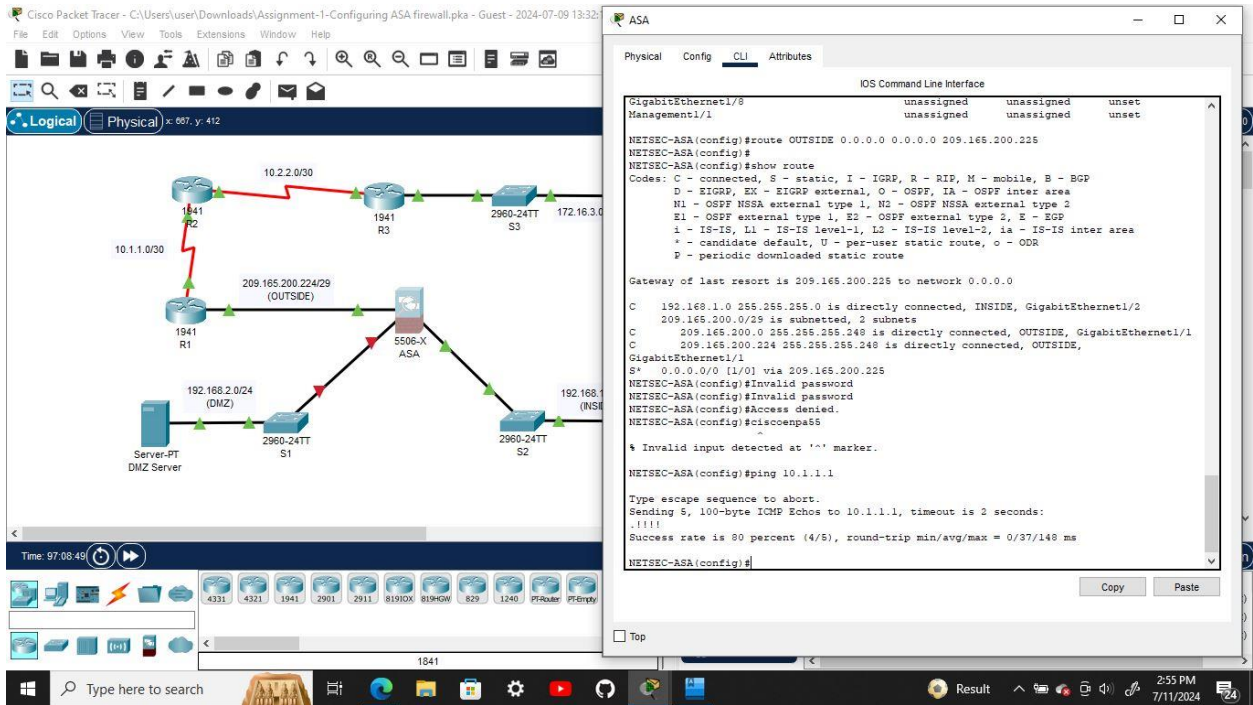
- Create a “quad zero” default route using the **route** command, associate it with the ASA OUTSIDE interface, and point to the R1 G0/0 IP address (209.165.200.225) as the gateway of last resort.



- b. Issue the **show route** command to verify the static default route is in the ASA routing table.

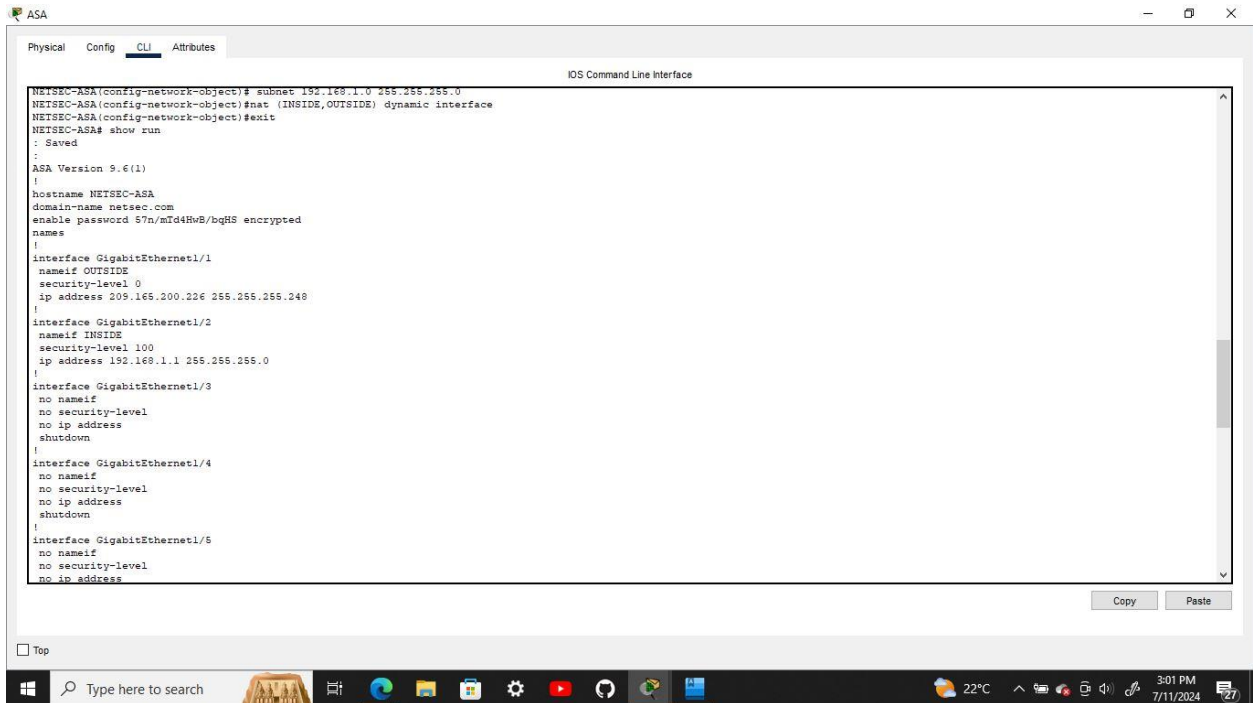


- c. Verify that the ASA can ping the R1 S0/0/0 IP address 10.1.1.1. If the ping is unsuccessful, troubleshoot as necessary.

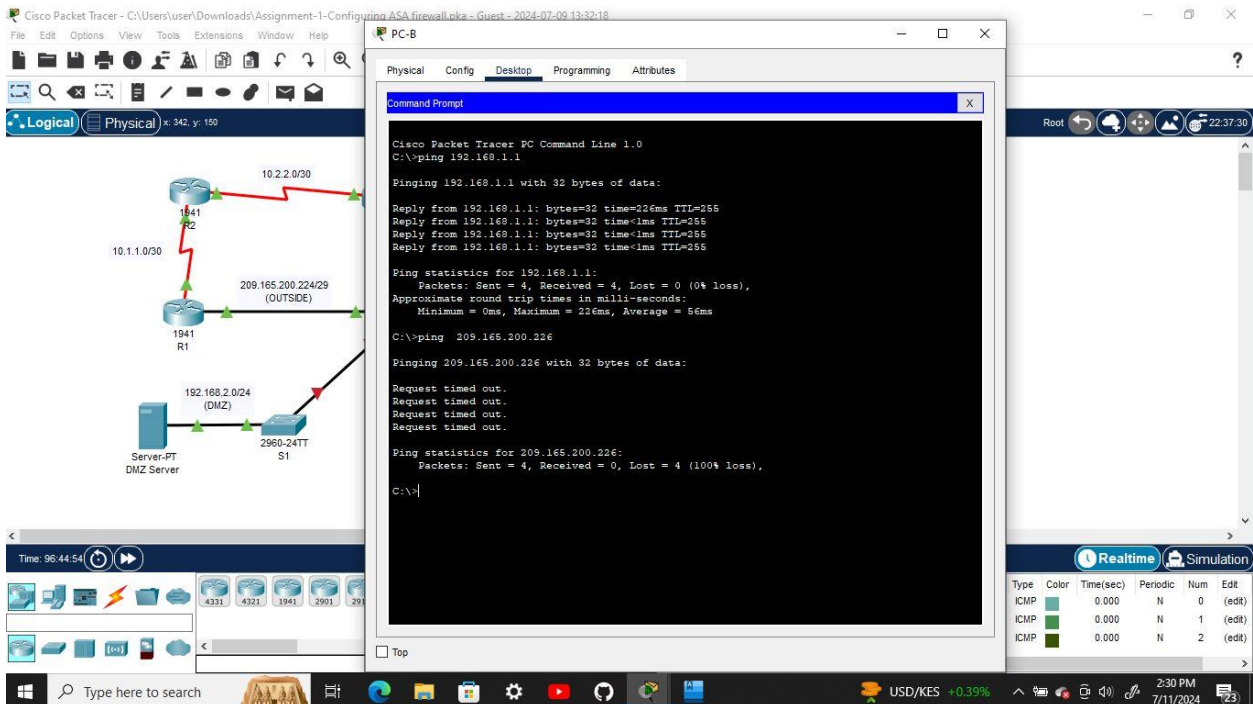


Step 2: Configure address translation using PAT and network objects.

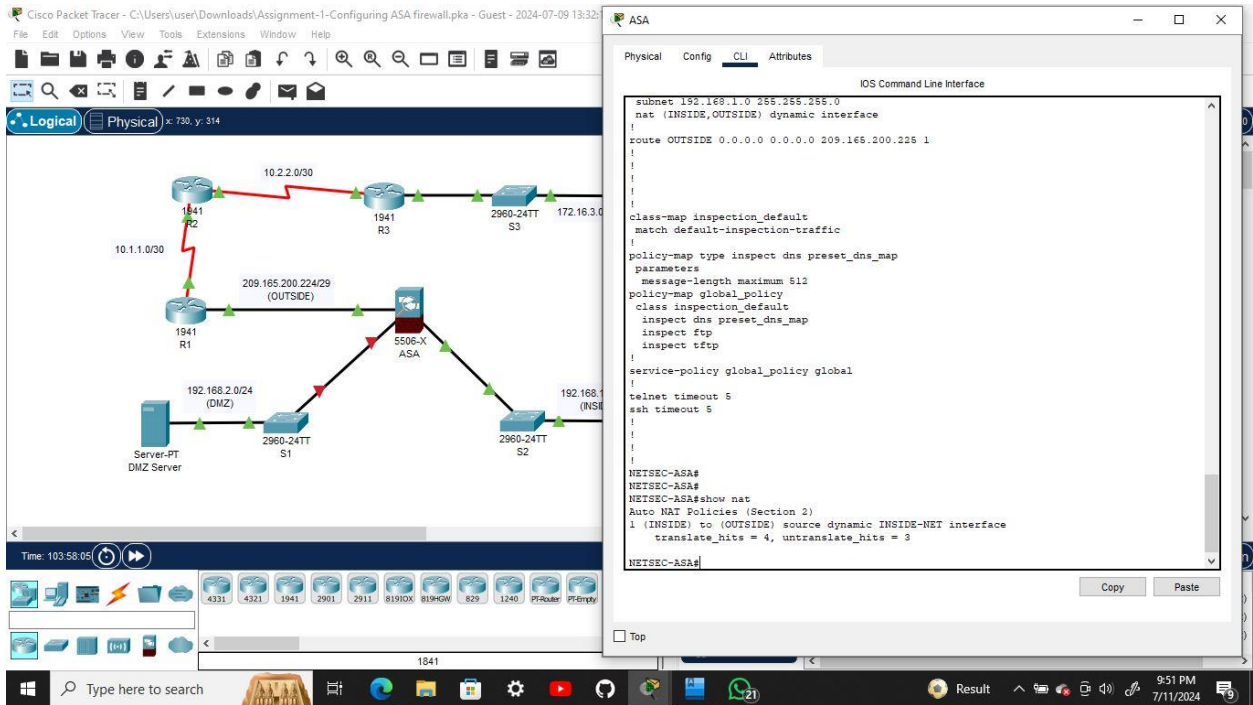
- Create network object **INSIDE-NET** and assign attributes to it using the **subnet** and **nat** commands.
- The ASA splits the configuration into the object portion that defines the network to be translated and the actual **nat** command parameters. These appear in two different places in the running configuration. Display the NAT object configuration using the **show run** command.



- c. From PC-B attempt to ping the R1 G0/0 interface at IP address 209.165.200.225. The pings should fail.



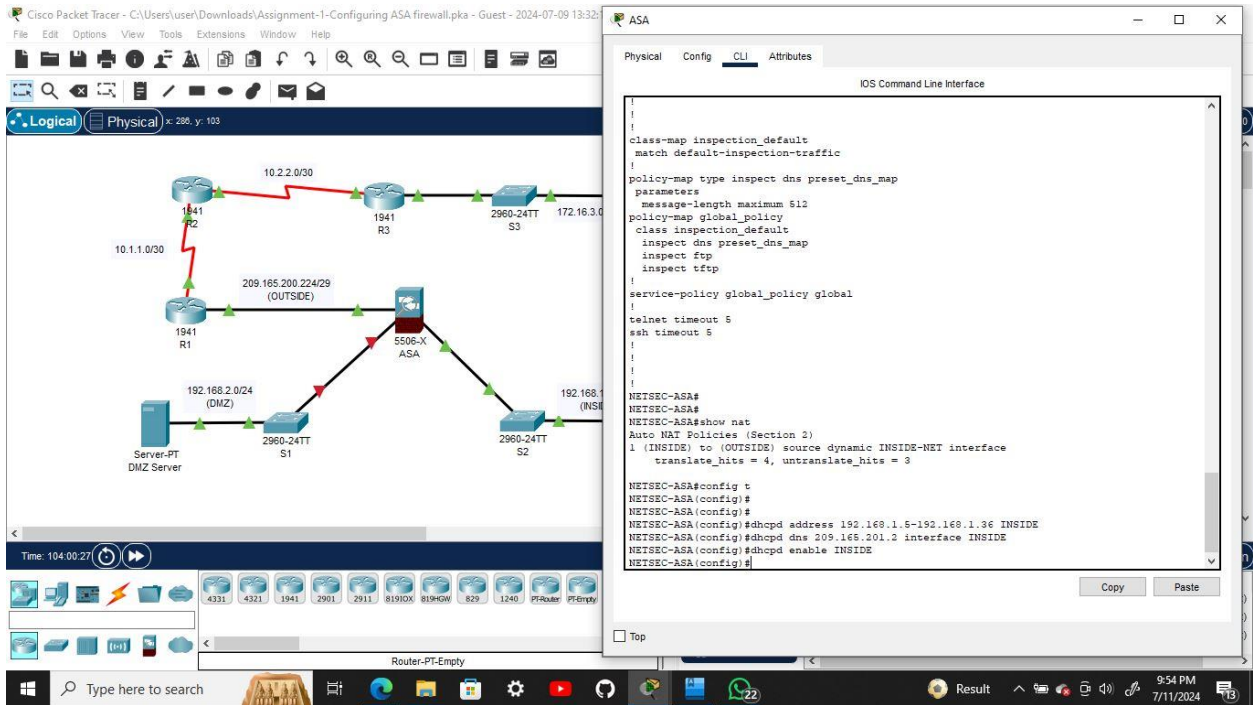
- d. Issue the **show nat** command on the ASA to see the translated and untranslated hits. Notice that, of the pings from PC-B, four were translated and four were not. The outgoing pings (echos) were translated and sent to the destination. The returning echo replies were blocked by the firewall policy.



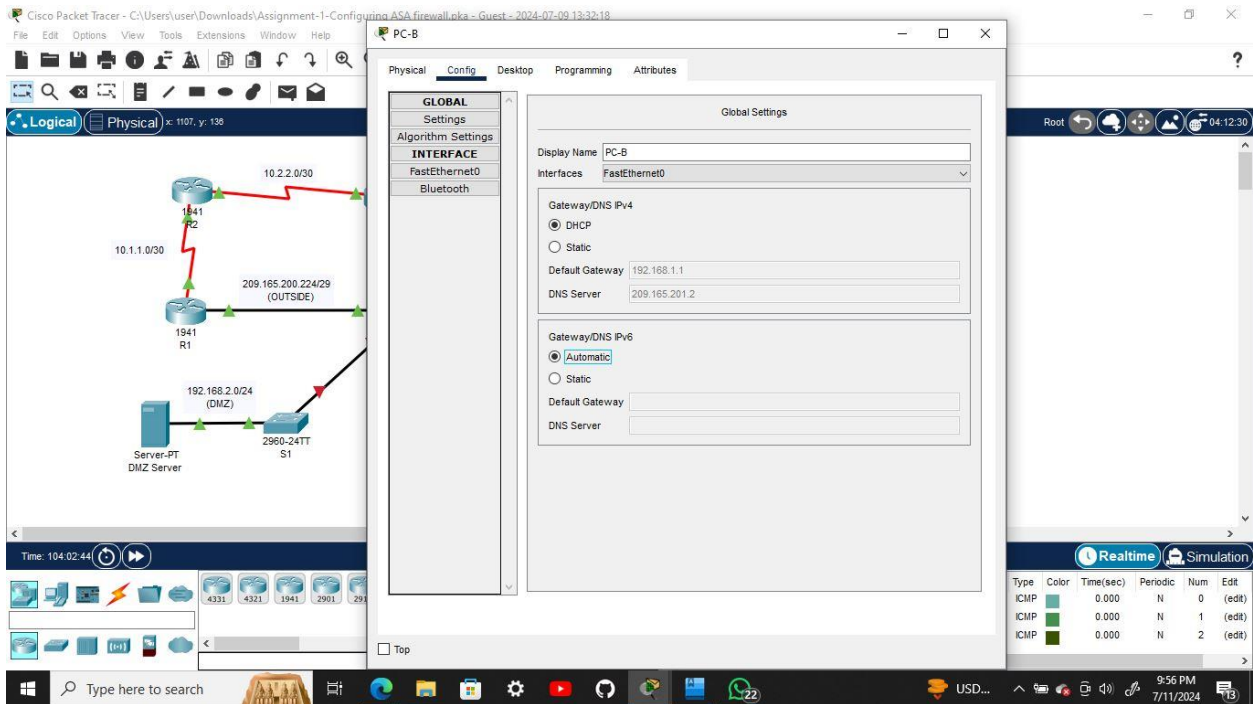
Part 4: Configure DHCP, AAA, and SSH

Step 1: Configure the ASA as a DHCP server.

- Configure a DHCP address pool and enable it on the ASA INSIDE interface.
- (Optional) Specify the IP address of the DNS server to be given to clients.
- Enable the DHCP daemon within the ASA to listen for DHCP client requests on the enabled interface (INSIDE).

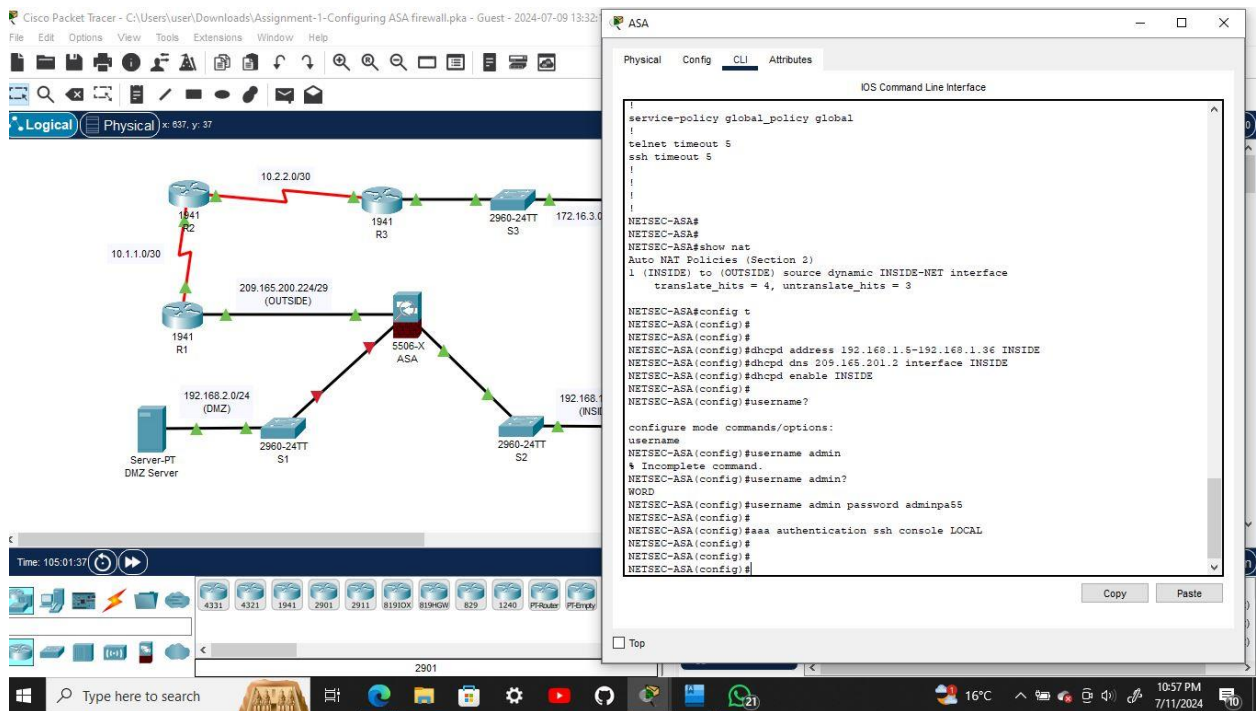


- d. Change PC-B from a static IP address to a DHCP client and verify that it receives IP addressing information. Troubleshoot, as necessary to resolve any problems.



Step 2: Configure AAA to use the local database for authentication.

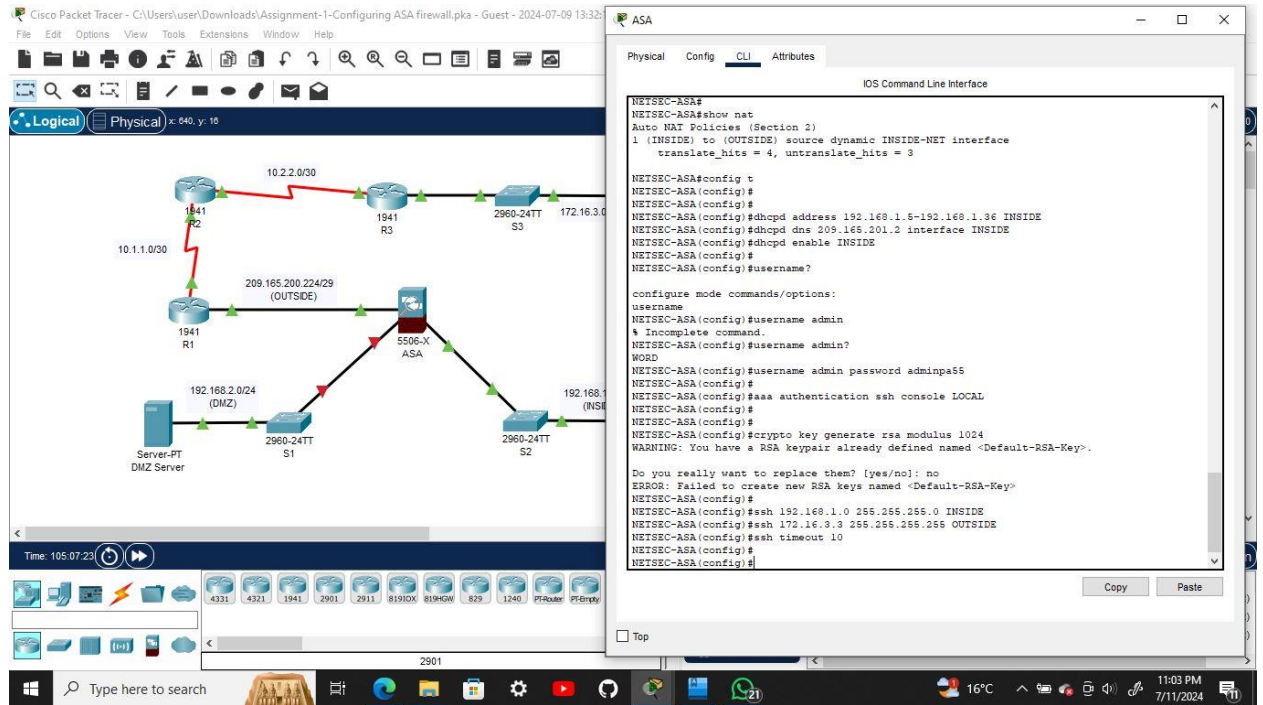
- Define a local user named **admin** by entering the **username** command. Specify a password of **adminpa55**.
- Configure AAA to use the local ASA database for SSH user authentication.



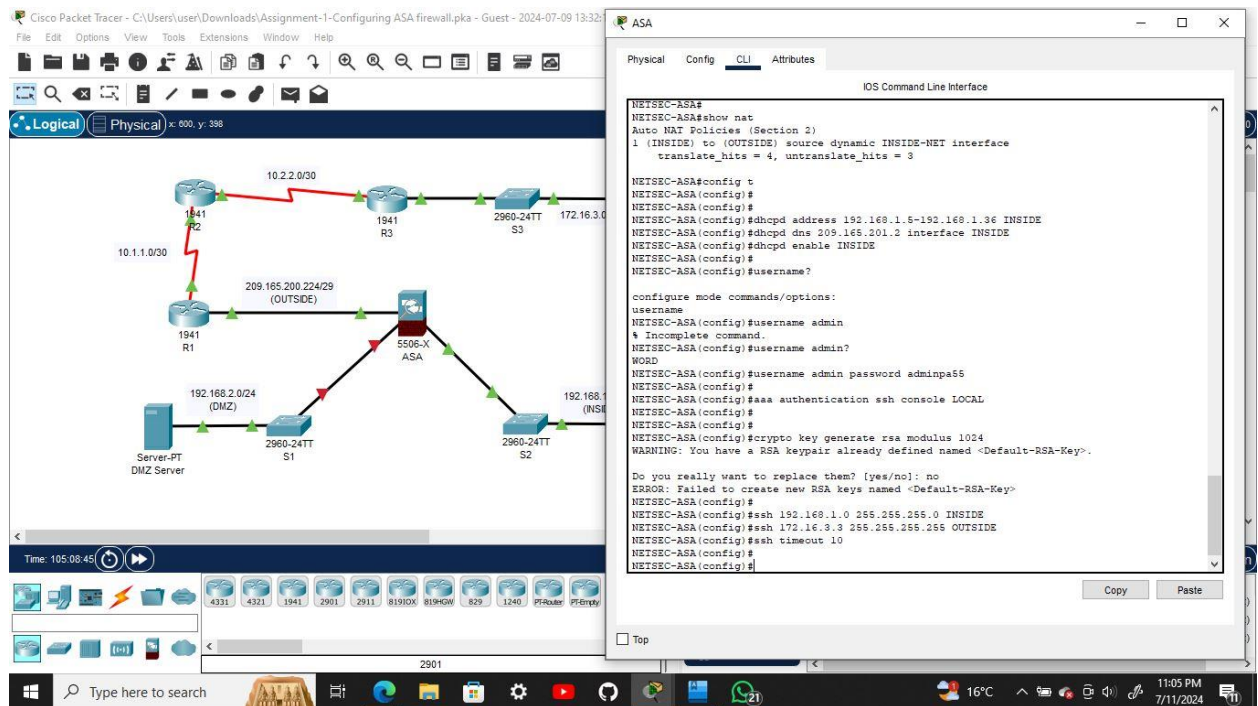
Step 3: Configure remote access to the ASA.

The ASA can be configured to accept connections from a single host or a range of hosts on the INSIDE or OUTSIDE network. In this step, hosts from the OUTSIDE network can only use SSH to communicate with the ASA. SSH sessions can be used to access the ASA from the inside network.

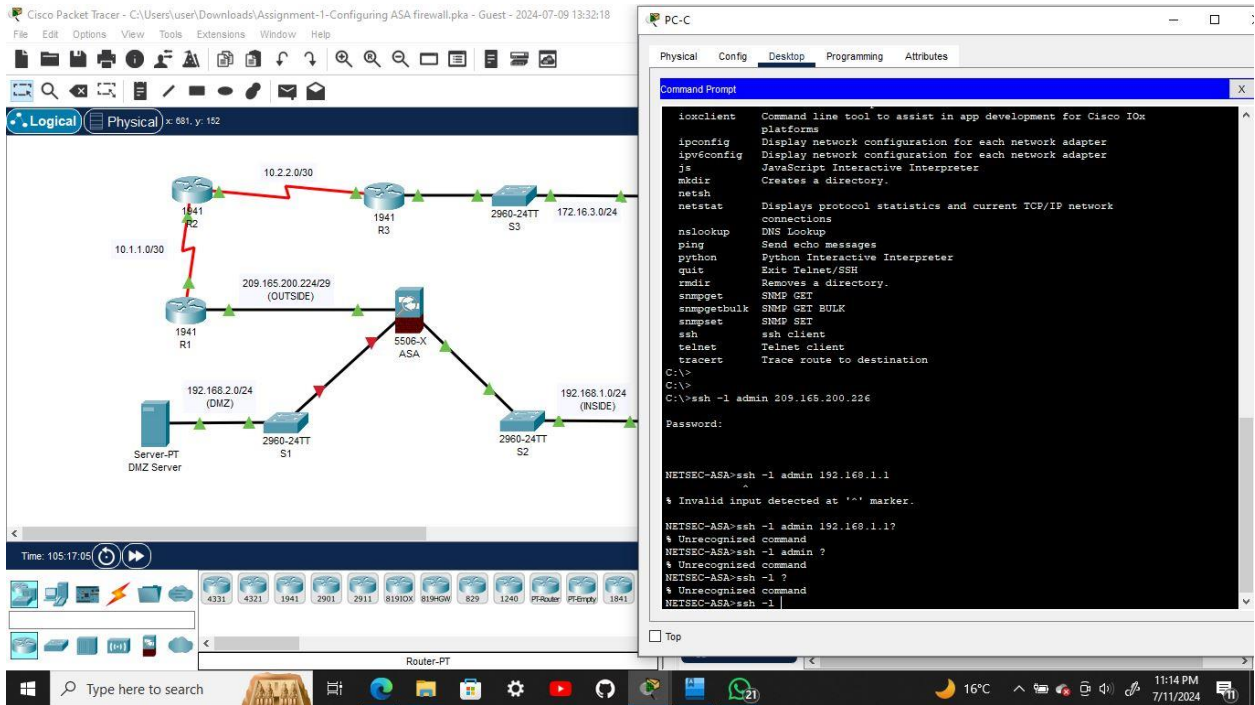
- Generate an RSA key pair, which is required to support SSH connections. Because the ASA device has RSA keys already in place, enter **no** when prompted to replace them.



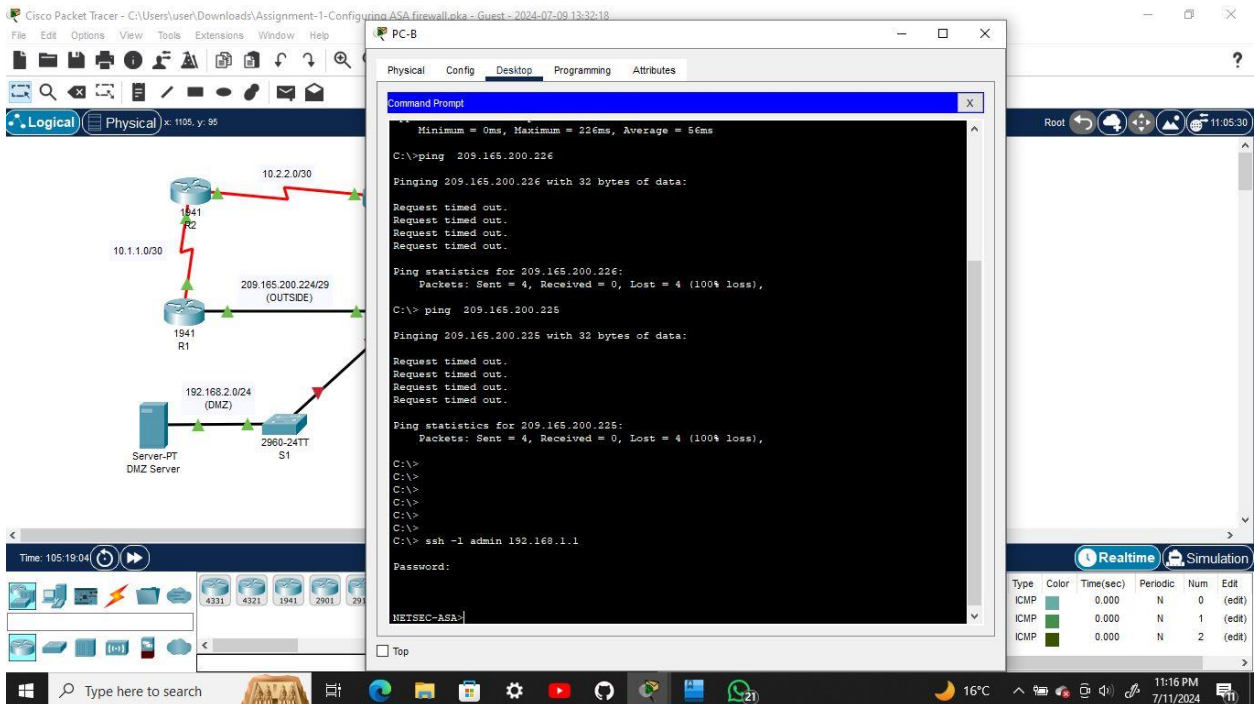
- b. Configure the ASA to allow SSH connections from any host on the INSIDE network (192.168.1.0/24) and from the remote management host at the branch office (172.16.3.3) on the OUTSIDE network. Set the SSH timeout to 10 minutes (the default is 5 minutes).



- c. Establish an SSH session from PC-C to the ASA (209.165.200.226). Troubleshoot if it is not successful.



- d. Establish an SSH session from PC-B to the ASA (192.168.1.1). Troubleshoot if it is not successful.



Part 5: Configure a DMZ, Static NAT, and ACLs

R1 G0/0 and the ASA OUTSIDE interface already use 209.165.200.225 and .226, respectively. You will use public address 209.165.200.227 and static NAT to provide address translation access to the server.

Step 1: Configure the DMZ interface VLAN 3 on the ASA.

- Configure DMZ VLAN 3, which is where the public access web server will reside. Assign it IP address 192.168.2.1/24, name it **DMZ**, and assign it a security level of 70. Because the server does not need to initiate communication with the inside users, disable forwarding to interface VLAN 1.

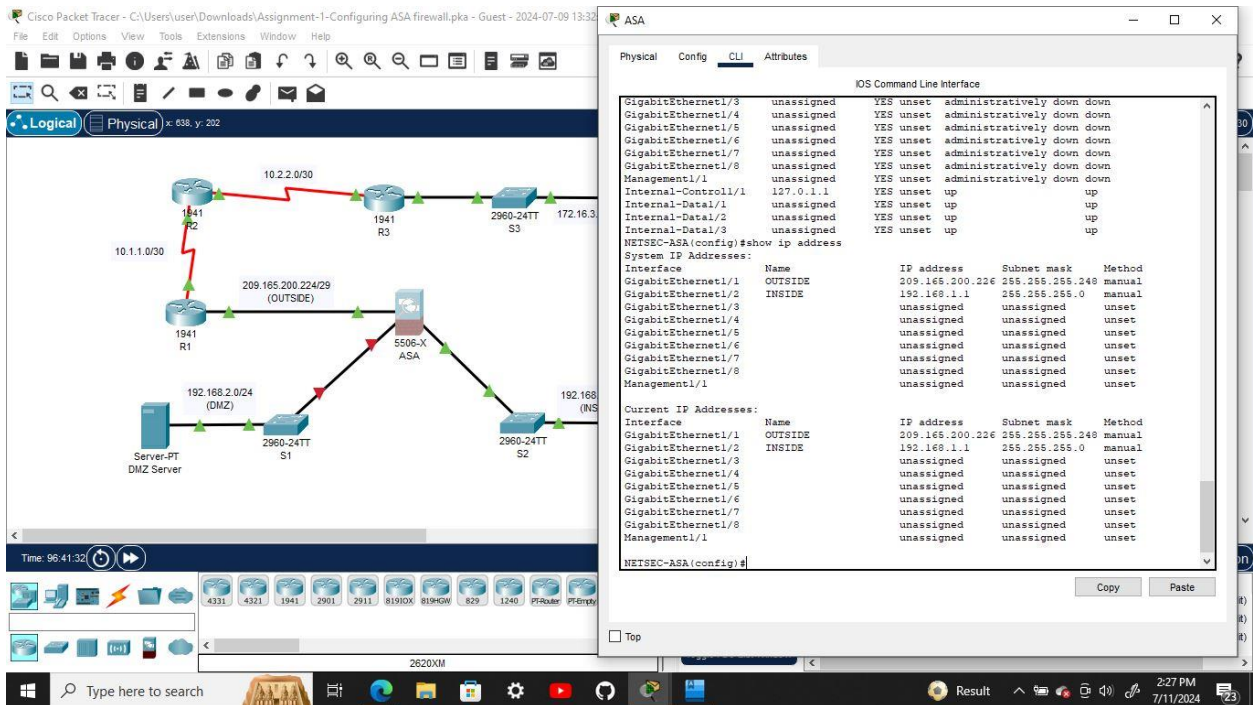
The image shows a Cisco Packet Tracer network diagram and the ASA configuration window. The network diagram includes a central ASA (5506-X) connected to three switches (S1, S2, S3) and a server (PT DMZ Server). The ASA is connected to S1 via Gig0/0 (OUTSIDE) and Gig1/1. S1 is connected to S2 via Gig0/24 (DMZ) and Gig1/1. S2 is connected to S3 via Gig0/24 (INSIDE) and Gig1/1. The ASA is also connected to S3 via Gig0/24 (INSIDE) and Gig1/1. The ASA configuration window shows the following commands:

```
NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#dhcpd address 192.168.1.5-192.168.1.36 INSIDE
NETSEC-ASA(config)#dhcpd dns 209.165.201.2 interface INSIDE
NETSEC-ASA(config)#dhcpd enable INSIDE
NETSEC-ASA(config)#
NETSEC-ASA(config)#username?
NETSEC-ASA(config)#
configure mode commands/options:
username
NETSEC-ASA(config)#username admin
% Incomplete command.
NETSEC-ASA(config)#username admin?
WORD
NETSEC-ASA(config)#username admin password adminp65
NETSEC-ASA(config)#
NETSEC-ASA(config)#aaa authentication ssh console LOCAL
NETSEC-ASA(config)#
NETSEC-ASA(config)#crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.
Do you really want to replace them? [yes/no]: no
ERROR: Failed to create new RSA keys named <Default-RSA-Key>
NETSEC-ASA(config)#
NETSEC-ASA(config)#ssh 192.168.1.0 255.255.255.0 INSIDE
NETSEC-ASA(config)#ssh 172.16.3.3 255.255.255.255 OUTSIDE
NETSEC-ASA(config)#ssh timeout 10
NETSEC-ASA(config)#
NETSEC-ASA(config)#interface gi1/3
NETSEC-ASA(config-if)#ip address 192.168.2.1 255.255.255.0
NETSEC-ASA(config-if)#nameif DMZ
INFO: Security level for "DMZ" set to 0 by default..
NETSEC-ASA(config-if)#security-level 70
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config-if)#
```

- Use the following verification commands to check your configurations:

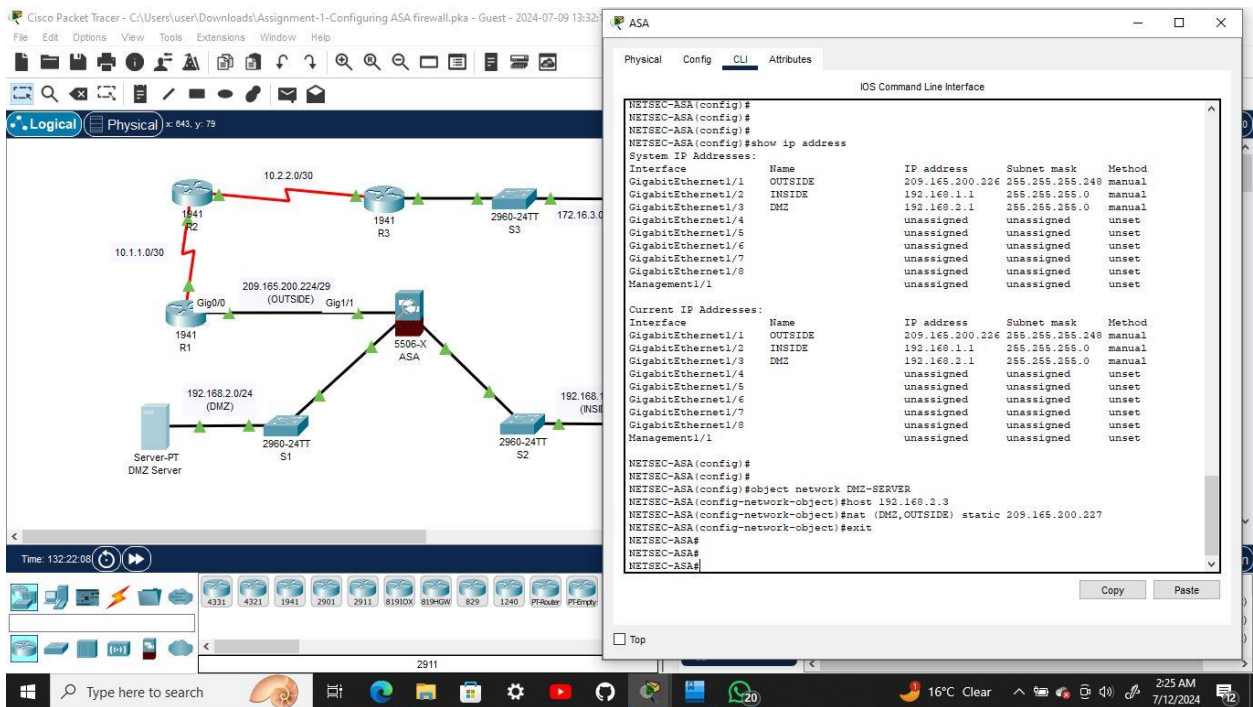
Use the **show interface ip brief** command to display the status for the ASA interfaces.

Use the **show ip address** command to display the information for the ASA interfaces.



Step 2: Configure static NAT to the DMZ server using a network object.

Configure a network object named **DMZ-SERVER** and assign it the static IP address of the DMZ server (192.168.2.3). While in object definition mode, use the **nat** command to specify that this object is used to translate a DMZ address to an OUTSIDE address using static NAT, and specify a public translated Subnet address of 209.165.200.227.



Step 3: Configure an ACL to allow access to the DMZ server from the Internet.

Configure a named access list **OUTSIDE-DMZ** that permits the TCP protocol on port 80 from any external host to the internal IP address of the DMZ server. Apply the access list to the ASA OUTSIDE interface in the "IN" direction.

The screenshot displays the Cisco Packet Tracer interface with a network diagram and the ASA configuration window. The network diagram shows a central ASA (5506-X) connected to three routers (R1, R2, R3) and two switches (S1, S2). R1 is connected to the DMZ server (209.165.200.227) via its Gig0/0 interface. R2 is connected to the Internet (10.1.1.0/30) via its Gig0/0 interface. R3 is connected to the Internet (10.2.2.0/30) via its Gig0/0 interface. S1 is connected to the DMZ server via its Gig1/0 interface. S2 is connected to the Internet (192.168.1.0/24) via its Gig1/0 interface. The ASA configuration window shows the following configuration:

```
ASA
System IP Addresses:
Interface Name IP address Subnet mask Method
GigabitEthernet1/1 OUTSIDE 209.165.200.226 255.255.255.248 manual
GigabitEthernet1/2 INSIDE 192.168.1.1 255.255.255.0 manual
GigabitEthernet1/3 DMZ 192.168.2.1 255.255.255.0 manual
GigabitEthernet1/4 unassigned unassigned unset
GigabitEthernet1/5 unassigned unassigned unset
GigabitEthernet1/6 unassigned unassigned unset
GigabitEthernet1/7 unassigned unassigned unset
GigabitEthernet1/8 unassigned unassigned unset
Management1/1 unassigned unassigned unset

Current IP Addresses:
Interface Name IP address Subnet mask Method
GigabitEthernet1/1 OUTSIDE 209.165.200.226 255.255.255.248 manual
GigabitEthernet1/2 INSIDE 192.168.1.1 255.255.255.0 manual
GigabitEthernet1/3 DMZ 192.168.2.1 255.255.255.0 manual
GigabitEthernet1/4 unassigned unassigned unset
GigabitEthernet1/5 unassigned unassigned unset
GigabitEthernet1/6 unassigned unassigned unset
GigabitEthernet1/7 unassigned unassigned unset
GigabitEthernet1/8 unassigned unassigned unset
Management1/1 unassigned unassigned unset

NETSEC-ASA(config)#
NETSEC-ASA(config)#
NETSEC-ASA(config)#object network DMZ-SERVER
NETSEC-ASA(config-network-object)#host 192.168.2.3
NETSEC-ASA(config-network-object)#nat (DMZ,OUTSIDE) static 209.165.200.227
NETSEC-ASA(config-network-object)#exit
NETSEC-ASA#
NETSEC-ASA#
NETSEC-ASA(config)#access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3
NETSEC-ASA(config)#access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq 80
NETSEC-ASA(config)#access-group OUTSIDE-DMZ in interface OUTSIDE
NETSEC-ASA(config)#
```

Step 4: Test access to the DMZ server.

From a web browser on PC-C, navigate to the DMZ server (209.165.200.227). Troubleshoot if it is not successful.

The screenshot displays the Cisco Packet Tracer interface with a network diagram and the PC-C configuration window. The network diagram is identical to the one in Step 3. The PC-C configuration window shows the following configuration:

```
PC-C
Physical Config Desktop Programming Attributes
Web Browser
URL http://209.165.200.227
Go Stop
```


Conclusion

This assignment was a learning experience. I got to know more about device configuration. The trouble I ran into while setting the clock also served to humble me as I lingered on it for some minutes. To make sure I got it right after the fifth frustrating try, I looked up how to go about it, and I was able to set the clock and date. It was something elementary but now I'm confident about my ability and will be sure to approach any assignment as someone who knows nothing and is there to learn. I encountered no further trouble and I was able to configure the devices, set up security and as a final step, checked to see the connectivity of all devices. They were all connected and running as expected to my relief.